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The altitude dependence of magnetic remanence in the slowly-cooled Precambrian plutonic terrain of West Greenland

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To evaluate the change in magnetic remanence with altitude through a slowly-cooled Precambrian basement terrain three vertical sections have been sampled in West Greenland. The study employs the principle that higher structural levels passed through their blocking temperatures earlier than lower levels and therefore record earlier pole positions, and it utilises dolerites and diorites intruded late in, or after, the tectonic history to minimise anisotropy effects. In the amphibolite facies terrain at Qâqatoqaq (1400 m) highly-stable magnetite-held remanences move on demagnetisation progressively along small circles interpreted to record younger to older apparent polar wander (a.p.w.) motions during cooling through the blocking temperature ranges. Although the raw data show no systematic variation with altitude, when account is taken of the blocking temperature spectra as defined by thermal demagnetisation there is a systematic change in palaeofield direction in the same sense as that recorded by the demagnetisation trends. Granulite facies terrain at Igdlúnguit qulât (600 m) again shows systematic variation with altitude when the sites are divided into those with a remanence dominated by hemo-ilmenite and those dominated by magnetite. A third section at Praestefjeldet (250 m) yields a palaeofield reversal and a high blocking temperature component.

The age evidence is evaluated to suggest that the a.p.w. path defined by 5 mean palaeopoles between 318°E, 1°N and 247°E, 38°N represents up to 50 Ma of palaeofield motion recorded by the uplift and cooling of this basement terrain at crustal depths of the order of 10 km. The calculated rate of a.p.w. motion is 1–2°/Ma and the rate of crustal uplift 10–20 m/Ma, these rates are respectively up to an order higher, and at least an order lower, than Phanerozoic rates. The collective data from Greenland agree closely with post-"Hudsonian" poles from the Laurentian Shield and represent part of a very widespread uplift event following this mobile episode. They show that altitude sections can yield a systematic record of the magnitude and direction of Precambrian a.p.w. motions provided that the blocking temperature spectra are taken into account.

1. Introduction

Very slow cooling during erosion and uplift has the resultant effect of lowering and broadening the blocking temperature range of the magnetic phases in rocks [1], and it is now widely recognised that a number of distinctive characteristics are associated with the magnetisations acquired in this way. Specifically, these characteristics [2] are: (1) secular variation is averaged out in individual samples, and results in very high values of the within-site precision parameter (k) if blocking temperature ranges are similar within a site, but low values if there is appreciable within-site variation; (2) all rocks formed before regional cooling will have comparable magnetisations regardless of their different modes and ages of origin provided that their blocking temperature ranges are similar; (3) if appreciable apparent polar wander (a.p.w.) occurred during cooling, samples from different structural levels should be distributed along the a.p.w. path; (4) samples from the same rock unit at higher crustal levels should be magnetised before those at lower crustal levels; (5) at any given crustal level samples with higher blocking temperatures should record earlier parts of the a.p.w. record than those with lower blocking temperatures; (6) in principle, progressive demagnetisation

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should recover progressively higher blocking temperature, and therefore older, segments of the total magnetisation, and sample directions will move back along the a.p.w. path.

Archaean and Proterozoic plutonic regions are ideal terrains for the study of uplift-related magnetisations because the rates of uplift were much slower than those of Phanerozoic orogenic belts [3]. In studies of the Kangamiut dykes exposed at sea level in Itividleqfjord, West Greenland, Morgan [2,4] clearly demonstrated effects (1), (2), (5) and (6) and was able to define a segment of the a.p.w. path at ca. 1700 Ma. To determine whether or not an altitude dependence of magnetic remanence (effects (3) and (4)) was demonstrable in this terrain, and reflected in a regional tilt, Beckmann et al. [5] sampled at sea level across a metamorphic facies boundary in Nordre Stromfjord and attempted to calibrate their study by reference to a vertical section of 620 m up the mountain Kaellingehaetten near the town of Holsteinsborg. Unfortunately, the rocks sampled by these authors possessed tectonic fabrics necessitating a variable, and sometimes large, correction for magnetic anisotropy. Such corrections must remain uncertain because the bulk anisotropy determined by conventional methods cannot be directly linked to the anisotropy of that fraction of the magnetic component possessing the stable magnetic remanence [6] (although TRM studies suggested that this correction was accurately applicable to the collection of Beckmann et al. [5]). Accordingly, the most satisfactory analysis of this phenomenon is likely to be achieved by restricting the study to rocks with little or no tectonic fabric. The studies reported here utilise igneous rocks intruded at a late stage, or after the sequence of tectonic events in this area and have minimal tectonic fabric. They employ three mountain sections in the West Greenland plutonic terrain; the two sections in granulite facies rocks achieved ca. 600 and 250 m thick sections and the one in amphibolite facies rocks is ca. 1400 m thick.

2. Geology

The NNE-trending Kangamiut dykes were intruded at 1950 $\pm$ 60 Ma [7] in between a high-strain (Nag 1) tectonic episode prior to 2500 Ma which imposed a prominent ENE tectonic fabric on Archaean terrain, and a more localised lower strain (Nag 2) episode. The Nag 2 episode deformed the Kangamiut dykes along a linear zone trending ENE and achieved overthrusting by simple shear to the SSE. In the areas of this study, however, the effects of Nag 2 episode were minor.

The Kangamiut dykes form part of a conjugate swarm [8] and the non-consistency of their cross cutting relationships with the E-W to WNW-ESE dykes running through Itivdleqfjord suggests that both sets are essentially of the same age. Further south the NNE dykes appear to consistently cut the E-W dykes [9] and may be somewhat younger. The change in strike from predominantly NNE to predominantly E in the Nagssugtoqidian belt was originally thought to indicate the progressive deformation but Watterson [10] found that the ENE strike was controlled by the (Nag 1) pre-dyke tectonic fabric because igneous features show that they have not been reoriented. Differences in direction and form of the dykes across the boundary are attributed to ductility contrasts in the country rocks which are thought to have been at relatively high temperatures during intrusion: the amphibolite facies rocks were more ductile than the granulite facies at the same temperature and permit development of pitch and swell (boudinage) forms contrasting with the typical parallel-sided dykes intruding the granulite facies gneisses [11].

The Qáqatoqaq mountain traverse in Fig. 1 utilises these E-W dykes and is entirely in amphibolite facies rocks. Korstgaard [12] notes that the change from amphibolite to granulite facies country rock in this area is diffuse and expressed only by the appearance of the hypersthene and an associated change in colour of the gneisses from grey to brown; he interprets this change as a prograde metamorphic transition of Archaean age which thus predates dyke intrusion and regional uplift.

The other two traverses (Fig. 1) are in granulite facies terrain north of the Nagssugtoqidian belt at Igdlúnguit qulât and Praestefjeldet. Unlike the previous study in this region [5], they utilise a suite of late to post-Nag 2 diorites [13] observed north

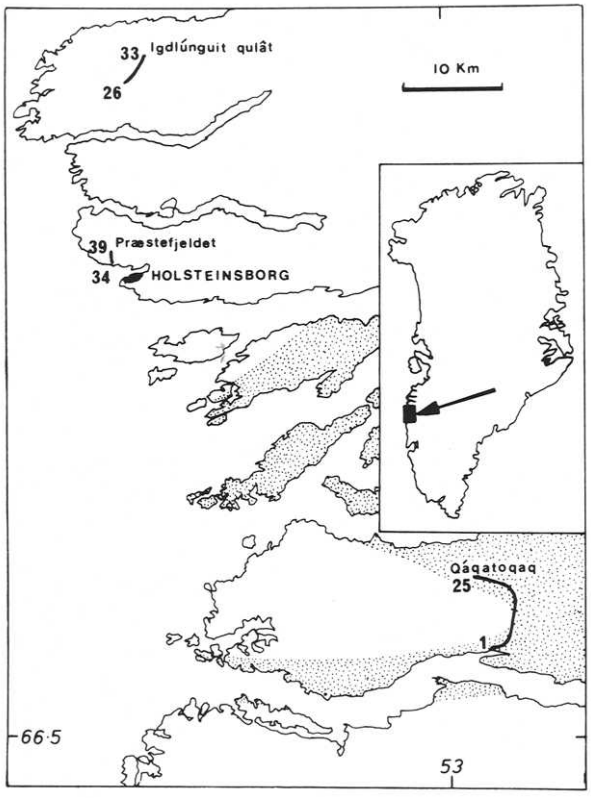


Fig. 1. The location of three palaeomagnetic studies reported here in the Precambrian plutonic terrain of West Greenland. Stippled areas are amphibolite facies rocks and blank areas are granulite facies rocks. In each case the lowest number is the site at the bottom of the section and the highest number is the site at the top of the section (see Table 1).

of 66°55′ which are largely unaffected by this episode with the exception of minor deformation at igneous contacts; they extensively intrude the terrain mostly as dykes but with a wide range of geometries. Hickman [14] has determined a preliminary 1700 Ma Rb-Sr age on these diorites in broad agreement with a Nag 2 pegmatite age of 1670 ± 150 Ma. Korstgaard [12] regards the granulite mineral assemblages in this area as Nag 2 in age, i.e. close to the diorite intrusion age and representing a maximum estimate of the magnetisation age.

3. The Palaeomagnetic studies

3.1. Sampling and laboratory procedure

A number (usually seven) of rock cores 2.4 cm diameter and up to 15 cm long were drilled using a portable motorised drill, and oriented in most cases by sun-compass. Magnetic declinations were also routinely recorded and in the few cases where a sun sight was not possible, orientation was made by magnetic compass in conjunction with topographic sights; altitudes were recorded with an aneroid calibrated each day. In the laboratory the cores were cut into 2.4 cm cylinders and at least one cylinder from each core was progressively demagnetised in alternating fields (a.f.), usually in steps of 5 or 10 milliteslas (mT), up to peak fields of 160 mT. One or more cores from each site were stepwise thermally demagnetised at temperatures or 150, 300, 400 and 500°C followed by increments of 30° or 20°C to the Curie point of magnetite (ca. 580°C) or of hematite (ca. 680°C). The behaviour of the magnetic moments with these progressive treatments is routinely expressed (Figs. 3 and 4) in terms of the change in direction as plotted on a stereonet and the decay of the initial (total) natural remanent magnetisation (NRM).

3.2. The Qáqatoqaq mountain section

This sample comprises three sites at sea level and 22 other sites at approximately equal intervals to the summit of the mountain (Table 1). All cores (with the exception of site 16) were drilled from basic dykes belonging to E-W members of the conjugate swarm [8]. Unaltered igneous textures are preserved in these rocks although autometamorphism has converted primary clinopyroxene to green hornblende [12], and thinner dykes are in general more altered than thick dykes and may contain garnet; these sites are thus all in regional amphibolite facies (Fig. 1).

Thermomagnetic determinations (saturation magnetisation J_s, versus temperature) identify a phase close to pure magnetite ($T_c \sim 580$°C) in composition but no hematite (Fig. 2 and Table 1) except at site 23. That magnetite is the principal,

TABLE 1

Palaeomagnetic results from sections through amphibolite and granulite facies basement terrains, central West Greenland

Site No.	Altitude (m)	N	D	I	R	k	Palaeomagnetic pole				Curie point(s)	J_0 ($\times 10^{-5}$ A m^2 kg^{-1})
							°N	°E	*dp*	*dm*		
(i) Qáqatoqaq section												
1	0	(i) 6	189.6	66.1	5.99	886	25.4	300.2	3.0	3.7	570	1.38–0.72
		(ii) 6	173.1	65.3	5.99	418	24,2	312.4	4.3	5.3		
2	0	(i) 6	190.3	66.4	5.96	122	25.8	299.7	8.2	10.0	565	1.18–0.54
		(ii) 6	175.2	63.6	5.94	81	21.9	310.9	9.4	11.9		
3	0	(i) 7	217.2	69.6	6.99	402	33.3	281.6	4.4	5.2	565	0.95–0.35
		(ii) 6	190.3	66.4	5.96	172	25.8	299.7	8.2	10.0		
4	40	7	194.9	68.1	6.99	1309	28.4	296.6	2.4	2.8	560	2.10–1.55
5	205	(i) 5	230.3	63.5	4.94	72	28.2	269.2	11.4	14.4	570	3.52–0.31
		(ii) 5	214.5	64.0	4.99	700	25.5	281.2	3.7	4.6		
6	278	(i) 7	234.4	61.7	6.98	309	27.2	265.2	4.1	5.3	560	13.67–2.59
		(ii) 7	223.9	59.9	6.96	143	22.6	272.5	5.8	7.6		
7	335	(i) 8	256.4	66.8	7.94	127	39.5	252.1	6.7	8.2	565	12.61–7.83
		(ii) 8	254.2	73.0	7.96	182	46.7	260.1	6.6	7.4		
8	275	(i) 6	257.9	61.1	5.98	276	33.8	246.6	4.8	6.2	560	29.22–3.91
		(ii) 6	261.4	67.8	5.99	356	42.4	249.4	4.9	5.9		
9	400	(i) 7	220.5	58.3	6.99	622	20.1	274.7	2.7	3.6	575	3.96–1.67
		(ii) 7	215.3	67.1	6.99	458	31.9	281.1	3.9	4.7		
10	445	(i) 6	218.9	54.3	5.80	25	18.9	274.3	13.7	19.4	570	0.91–0.32
		(ii) 6	223.7	64.6	5.97	180	30.1	273.9	6.4	8.0		
11	550	(i) 7	188.2	65.3	6.97	210	26.9	301.0	5.5	6.8	575	1.31–0.42
		(ii) 7	166.2	62.9	6.94	97	24.1	317.9	7.6	9.7		
12	625	(i) 7	211.6	65.4	6.95	112	29.0	283.3	7.6	9.3	575	1.32–0.11
		(ii) 6	196.3	66.1	5.92	65	21.1	295.7	11.2	13.7		
13	682	(i) 7	193.2	65.0	6.97	184	19	297.7	5.8	7.2	575	2.24–0.37
		(ii) 7	178.3	62.5	6.92	74	16	308.5	8.6	11.1		
14	747	(i) 7	239.4	65.9	6.99	496	30	265.7	3.6	4.4	560	4.09–0.94
		(ii) 7	238.1	66.1	6.99	450	29.9	266.7	3.8	4.7		
15	820	(i) 7	224.1	65.9	6.99	520	24.7	274.9	3.5	4.3	575	3.60–2.29
		(ii) 7	212.1	75.4	6.99	4.78	41.3	288.1	4.7	5.1		
*16	820	(i) 7	73.9	60.0	6.60	15	43.2	32.0	18.4	24.4	575	0.23–0.05
		(ii) 7	77.9	24.8	6.58	14	16.7	43.2	9.5	17.8		
17	860	(i) 7	268.4	71.4	6.93	84	49.1	248.7	10.2	11.6	570	6.89–1.36
		(ii) 6	292.2	77.8	5.96	141	64.5	248.9	10.10	10.6		

18	880		4	167.0	62.7	3.99	1095	21.2	317.2	3.4	4.4	565	0.50–0.05
19	925	(i)	7	218.4	59.6	6.96	167	21.1	276.8	5.3	7.1	570	5.23–1.69
		(ii)	6	225.5	63.9	5.97	172	27.5	272.9	6.5	8.1		
20	995		5	196.9	71.7	4.97	155	33.8	296.1	9.5	10.9	560	0.09–0.02
21	1075		7	219.6	58.0	6.96	155	19.7	275.3	5.3	7.2	570	2.52–0.33
22	1195		7	200.6	54.5	6.91	64	13.0	290.0	7.5	10.7	560	8.11–2.86
23	1240		7	231.8	56.7	6.88	49	21.3	264.9	9.2	12.0	560, 650	6.83–0.56
24	1290	(i)	5	275.4	53.1	4.99	2057	32.7	227.3	1.6	2.3	580	36.88–34.93
		(ii)	5	261.1	52.4	4.99	812	26.6	239.2	2.5	3.7		
25	1375	(i)	7	85.2	71.8	6.85	39	31.8	9.4	15.2	17.3	565	0.13–0.04
		(ii)	3	243.0	64.1	2.94	31	32.3	254.9	28.6	35.8		
(ii) Igdlúnguit qulât section													
26 (10)	50		8	222.4	72.3	7.91	76.0	38.4	278.9	10.0	11.3	575, 650	5.65–0.48
27 (9)	125		7	213.2	71.9	6.93	81.7	36.4	284.6	10.4	11.9	570	1.72–1.10
28 (8)	230		9	194.7	71.0	8.93	112.3	32.9	296.6	7.4	8.5	565, 580	2.61–0.65
29 (7)	340		7	234.2	76.2	6.95	124.2	46.6	275.2	9.3	10.1	570, 680	1.37–0.53
30 (6)	375	(i)	2	172.3	47.6	1.99	177.2	5.9	313.2	15.9	24.5	570, 670	2.83–0.65
(5)		(ii)	4	314.0	75.1	3.91	33.4	70.0	223.1	26.9	29.5		
31 (4)	470		10	208.6	63.5	9.76	36.8	24.2	284.7	10.1	12.8	570	1.16–0.11
32 (3)	560	(i)	5	165.9	39.2	4.92	51.5	−6.2	319.5	7.7	12.9	520, 660	1.67–0.26
(2)		(ii)	6	167.4	35.5	5.91	56.6	−2.8	318.3	6.0	10.4		
33 (1)	615		6	196.8	61.6	7.62	18.3	20.4	294.9	15.9	20.6	555	1.39–0.45
(iii) Praestefjeldet section													
34 (7)	210	(i)	5	62.1	−15.6	4.98	184.0	3.1	65.1	2.9	5.8	550, 680	4.14–0.77
(6)		(ii)	5	250.7	25.9	4.92	44.7	5.3	239.3	6.4	11.8		
35 (5)	345		4	64.9	−13.4	3.99	347.6	3.2	62.0	2.6	5.0	560	1.61–1.41
36 (4)	380		9	81.6	−28.2	8.96	199.1	−10.6	49.9	2.2	4.0	570	10.60–1.20
37 (3)	485		7	84.7	−45.3	6.96	163.8	−22.5	51.9	3.8	6.0	550, 680	3.66–0.98
38 (2)	540		3	94.4	−28.6	2.77	8.8	−15.7	38.1	26.7	48.7	570	38.36–0.44
**39 (1)	535		4	197.8	63.3	3.99	504.1	22.6	292.4	5.1	6.5	560	2.10–0.36

* Amphibolite, remaining sites in the Qáqatoqaq section are dolerite dykes.

** Granulite-facies gneiss, remaining sites in the Igdlúnguit qulât and Praesfjeldet section are in late Nag 2 diorites.

When two site means are listed they represent the beginning and the end of small circle migration during treatment.

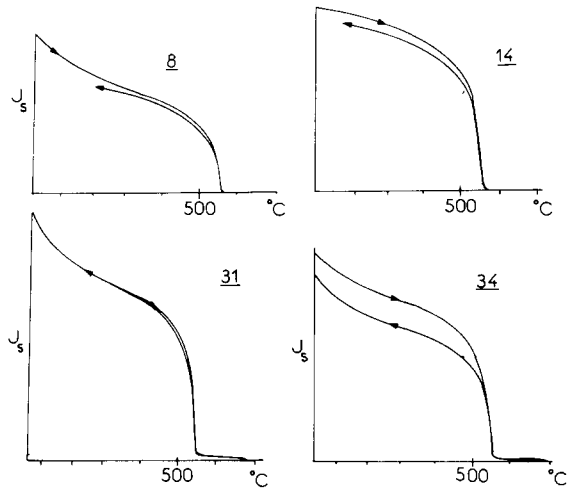


Fig. 2. Some representative thermomagnetic determinations from the granulite and amphibolite facies rocks of this study. Numbers are site numbers and the J_s axis is arbitrary.

or only, magnetisation carrier is confirmed by the exponential decrease in NRM with progressive a.f. treatment (15] and Fig 3), and the disappearance of coherent magnetisations (with the exception of sites 1 and 23) in the range 560–600°C with thermal treatment (Fig. 4) Morgan [4] identifies between 1 and 12% of opaque phases in the dykes of this area the bulk of which are homogeneous magnetite grains; mineral separation studies, however, have shown (G.E. Morgan, personal communication) that only the lowest coercivity remanence resides in these grains with the highest coercivity components being associated with the silicates.

The critical changes in direction of magnetisation with progressive treatment are identical to those observed by Morgan [2,4] and comprise (in many cores) a movement away from a steep positive direction associated with a low coercivity component, and usually completed within the first 5 mT of treatment, followed by minor but continuous and systematic movement along the arc of a small circle and predominantly represented by a decreasing declination. Thermal demagnetisation results (e.g. Fig. 4) are essentially comparable to the a.f. results although the low coercivity component is in some instances not removed until 400–500°C with the bulk of this decreasing declination concentrated in the critical blocking temperature range at ca. 530–580°C; thus for this particular collection the a.f. method is more discriminating and this generally proves to be the case for magnetite-bearing amphibolite facies rocks [4,16,17].

The relaxation equation for single-domain grains is [18]:

$$\frac{1}{\tau} = C_{\exp}\left(\frac{-vH_cJ_s}{2kT}\right)$$

and a particular blocking temperature may be contributed to by high coercivity, small-volume grains and low coercivity, large-volume grains. Thus the efficiency of a.f. demagnetisation in recovering higher blocking temperature fractions of the magnetisation will depend on the grain size and the shape distributions. In these rocks Morgan [2,4] interprets the systematic migratory motion with a.f. treatment to be recovering progressively higher blocking temperature components, and therefore to be moving back along the path of palaeofield migration. This is probably a valid interpretation in these rocks because the high coercivity remanence originates in very fine exsolved magnetite in the silicate phases. This explanation would accord with the continuous migration of NRM direction over the whole range of a.f. treatment and account for the predominant anticlockwise trend (Fig. 3). However, some insight into the complexity of the Precambrian a.p.w. motions recorded by these rocks is shown by the variations within a few individual sites: different cores sometimes exhibit movements in different directions (site 13) while other cores may show no significant changes with cleaning although more than one direction is recorded at the site (sites 23 and 25, Fig. 3). As a result, while some within-site precisions (k) are outstandingly high, others are not so because the sample directions are elongated (23) or bimodal (25) in some way. Site 16 is the only one in amphibolite country rock and has contrasting directional properties to the remaining sites. Total NRMs are distributed (Fig. 3) and migrate to end points which are also distributed; this might represent subtraction of a recent field component but the end points are not interpretable: although these could be transitional directions no evidence has been found for a field reversal at the remaining sites. In the elongate distribution at

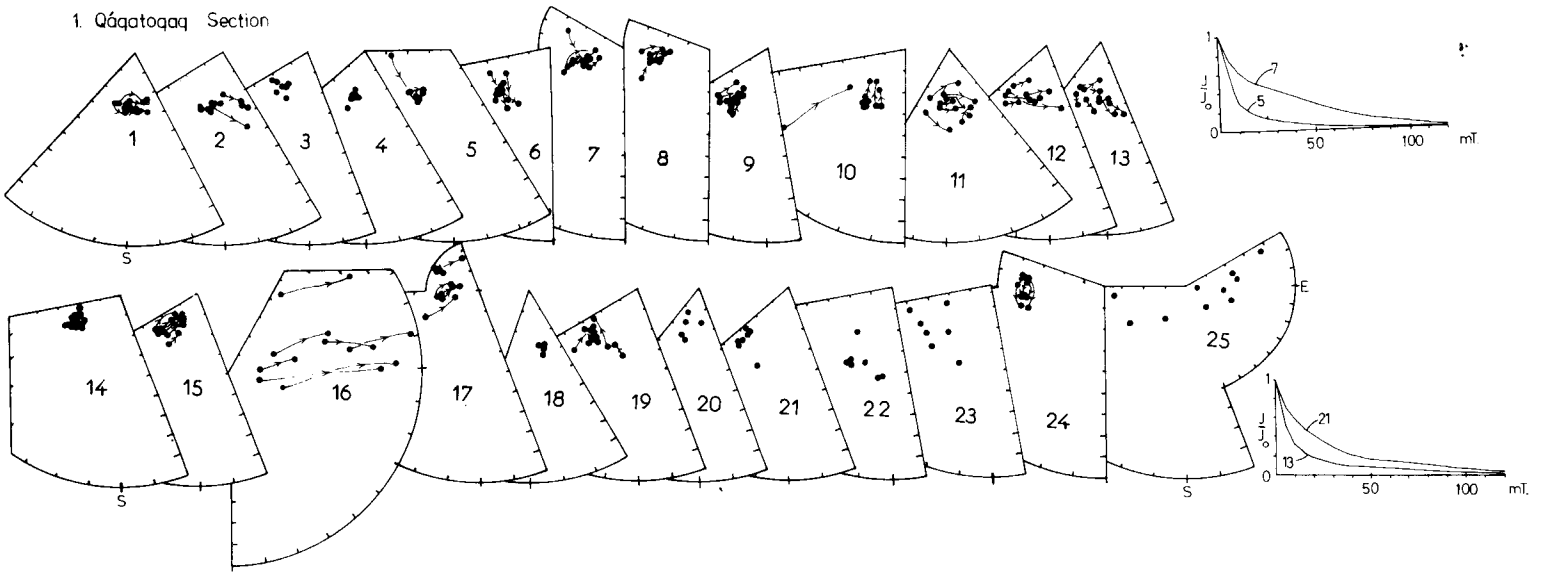


Fig. 3. The distribution of sample cleaned directions of magnetisation up the Qáqatoqaq section in order of increasing altitude. The directions are plotted after subtraction of any present field components in low fields. Where a continuous migration of the direction of magnetisation takes place over the range of cleaning treatment [2] the beginning and end point of this migration are plotted together with the direction of motion; note that (with the exception of country rock site 16) this motion is not away from the present field. Some representative a.f. demagnetisation curves expressing the fall off in remanence as a fraction of the initial moment, are also shown. Closed symbols are lower hemisphere plots and open symbols are upper hemisphere plots.

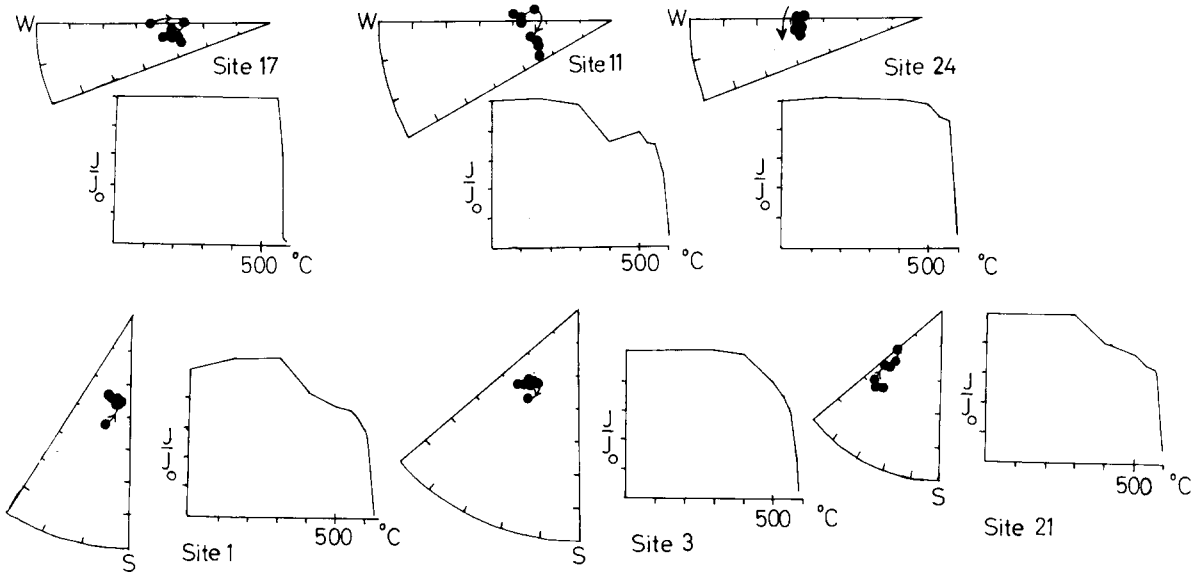


Fig. 4. Some examples of the behaviour of directions and intensities of magnetisation with progressive thermal cleaning.

site 25 the only clear difference is in the shape of the demagnetisation curves with the easterly positive group exhibiting a more rapid initial fall in moment.

With these minor exceptions the remaining sites exhibit highly systematic behaviour. In Table 1 and Fig. 6 site mean directions and palaeopoles are calculated and plotted at the beginning (usually after 5 or 10 mT of treatment) and end (either at 150 mT or when directional behaviour begins to become non-systematic) of the small circle migration. When the sample directions are plotted as a function of altitude up the section there are seen to be no systematic changes in direction of magnetisation recorded in these rocks (Fig. 3) and this result is similar to that of Beckmann et al. [5]. The collective data show that uplift of this ca. 1400 m of terrain was sufficient rapid not to have recorded any gross changes in the ambient geomagnetic field direction. However, when the more subtle changes in blocking temperature between sites are taken into account a much more systematic picture emerges. The thermal demagnetisation curves (e.g. Fig. 4) are generally considered to indicate the spectrum of blocking temperatures present in the rock although they will not record the actual blocking temperatures under conditions of slow cooling (section 2.5). Accordingly, we may estimate whether a magnetic component was acquired early or late in the magnetisation history by determining whether it resides predominantly in the lower or higher blocking temperature fraction. In Fig. 5 site palaeopoles (essentially a long time-average in these slowly-cooled rocks [2]) are differentiated on the basis of those sites showing more than 50% of the total NRM demagnetised below 550°C and those sites showing less than 50% of the total NRM demagnetised by 550°C. It is clear that a systematic a.p.w. motion from SE to NW is recorded firstly by the migration from high coercivity directions to the low coercivity directions (the anticlockwise migration identified during progressive cleaning) and secondly, with few exceptions (notably sites 17 and 18), by the distribution of the higher and lower coercive force spectra. This analysis also permits a systematic interpretation to be placed on the acquisition of magnetisation with depth in the crust as it cools from higher to lower levels: in the inset diagram the palaeopoles are connected in altitude sequence of those sites with more than 50% of the total NRM demagnetised between 500 and 550°C, between 555 and 580°C and above 580°C. The sequence of palaeopoles in each case confirms the a.p.w. mo-

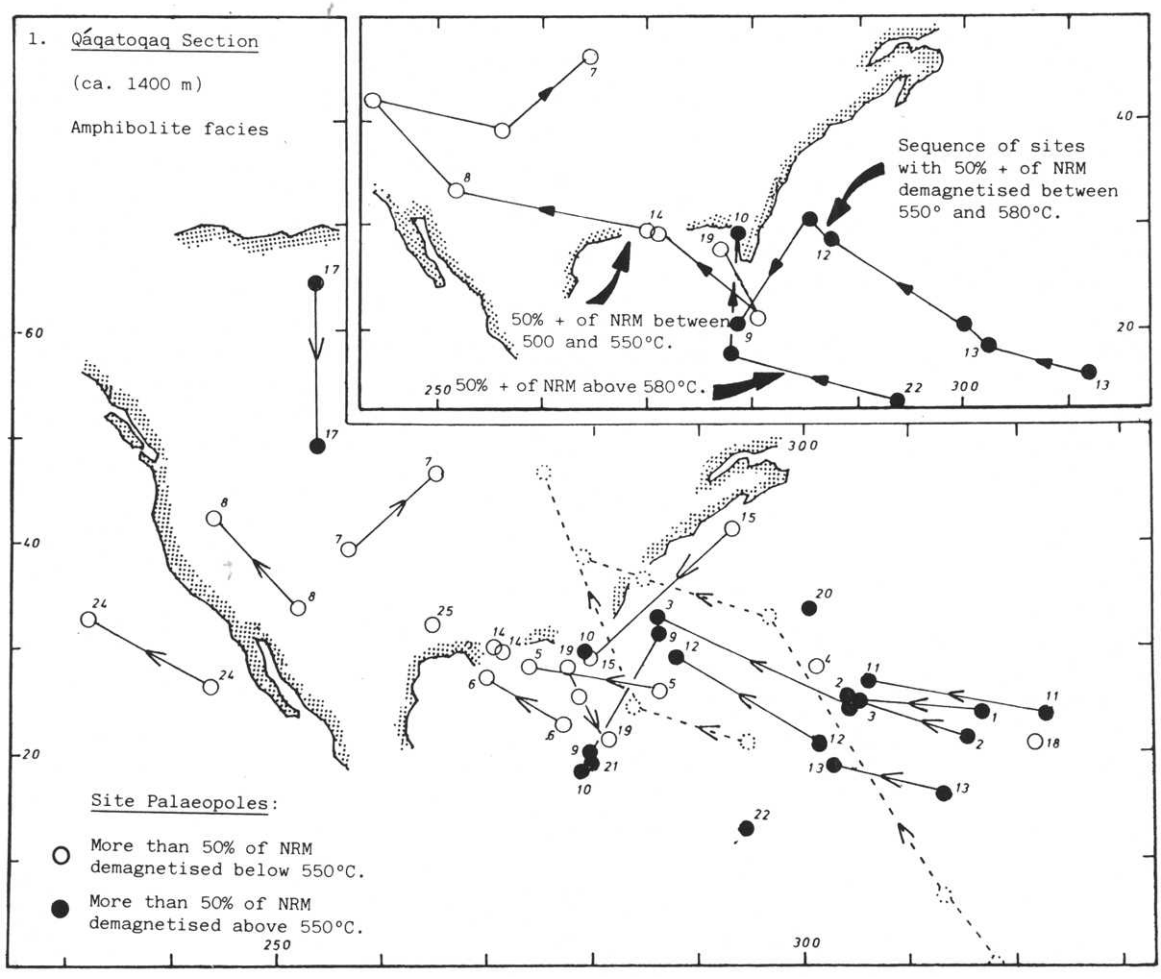


Fig. 5. The distribution of site palaeopoles from section 1 (Qáqatoqaq). The sites have been subdivided on the basis of thermal demagnetisation characteristics of one or more samples from the site, and the altitude sequence of sites with different blocking temperature properties (subdivided on a 50% basis) is shown in the inset diagram. The directions of apparent polar movement as indicated by the demagnetisation characteristics are indicated by the arrows.

tion identified by the other two methods (within-sample movements and high and low blocking temperature distributions), and show that altitude sections are a meaningful way of studying this variation if blocking temperature differences are taken into account.

Inevitably, the analysis is imperfect in this form because the thermal characteristics of a number of pilots are utilised to categorise poles based on a.f. characteristics; the varying a.f. demagnetisation characteristics of different samples from the same site show that this is not entirely valid. Also, the observation that not all sites in this amphibolite facies terrain fit a simple path indicates either that this analysis is inadequate or that a simple cooling explanation is inapplicable. The results from subsequent sections in granulite facies terrain where the blocking temperature spectra are narrower suggest that the former is the case, but this cannot be established with certainty because cooling conditions in the granulite and amphibolite facies terrains might have been radically different.

3.3. Igdlúnguit qulât section

The remaining sections in granulite facies rocks are dominated by a magnetic mineralogy contrast-

ing with the amphibolite facies terrains at Qáqatoqaq. The majority of cores have a remanence which is held wholly or partially by hematite or hemo-ilmenite; magnetic moments are frequently unaffected by a.f. demagnetisation (Fig. 6) above 15 mT and only fall to zero close to the Curie point of pure hematite with thermal demagnetisation (Figs. 7 and 8). The magnetite fraction dominates the thermomagnetic curves (Fig. 2) because magnetite has a much higher saturation remanence than hematite, although the latter is usually detected as a tail in these curves (Table 1).

Because the influence of the a.f. method is limited and because the thermal demagnetisation is less discriminating, there is no general tendency for the remanence directions to migrate along a small circle (sometimes following removal of a viscous component in low fields); only at site 32 was this behaviour detected and was then only a motion of 4° over the range of treatment. The distribution of sample mean directions is however sometimes elongate (sites 26, 27 and 32, Fig. 6) or bimodal (site 30) suggesting that a range of palaeofield components have been recorded by the highest blocking temperature fractions. In addition the trend of palaeopoles with altitude is largely systematic (Fig. 7) with the poles falling in a consistent trend along the polar path with decreasing altitude, and defining a trend that closely matches the high to low blocking temperature sequence identified in the amphibolite facies rocks of section 1 (Fig. 5). The sites falling on this trend (2, 3, 6, 8, 9 and 10 in Fig. 7, sites are renumbered as in Table 1) have step-shaped thermal demagnetisation curves with 50% or more of the total remanence remaining blocked above the Curie point of magnetite. The three sites (1, 4 and 7) which do not fit this trend define a subsidiary path (Fig. 7) and have contrasting thermal demagnetisation curves: only a small fraction comprising 16% or less of the initial remanence remains blocked above the Curie point of magnetite and no coherent directions of magnetisation were recovered above 600°C. At site 30 where there is a bimodal distribution of directions, the second group (5) with NW positive directions (Fig. 6) are magnetite-held and have low coercivity; they may comprise an extension of the group $1 \rightarrow 4 \rightarrow 7$. These two alti-

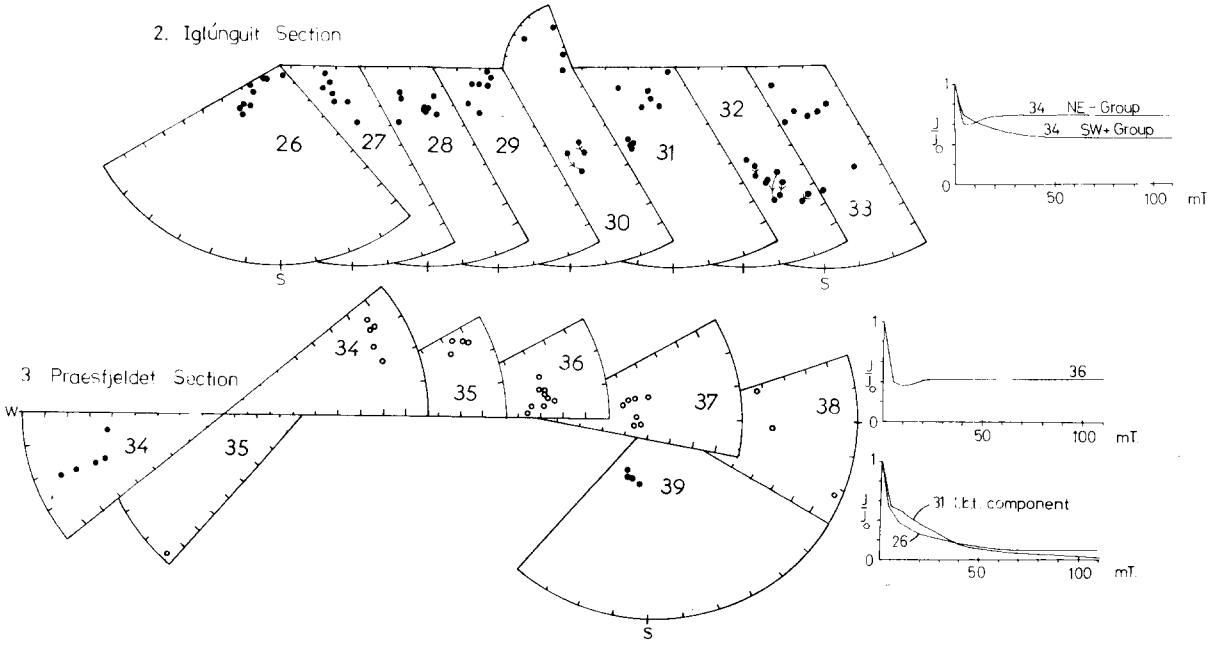


Fig. 6. The distribution of sample cleaned directions of magnetisation in section 2 and 3 in order of increasing altitude. Some representative demagnetisation curves are also shown. Symbols, etc., are as for Fig. 2.

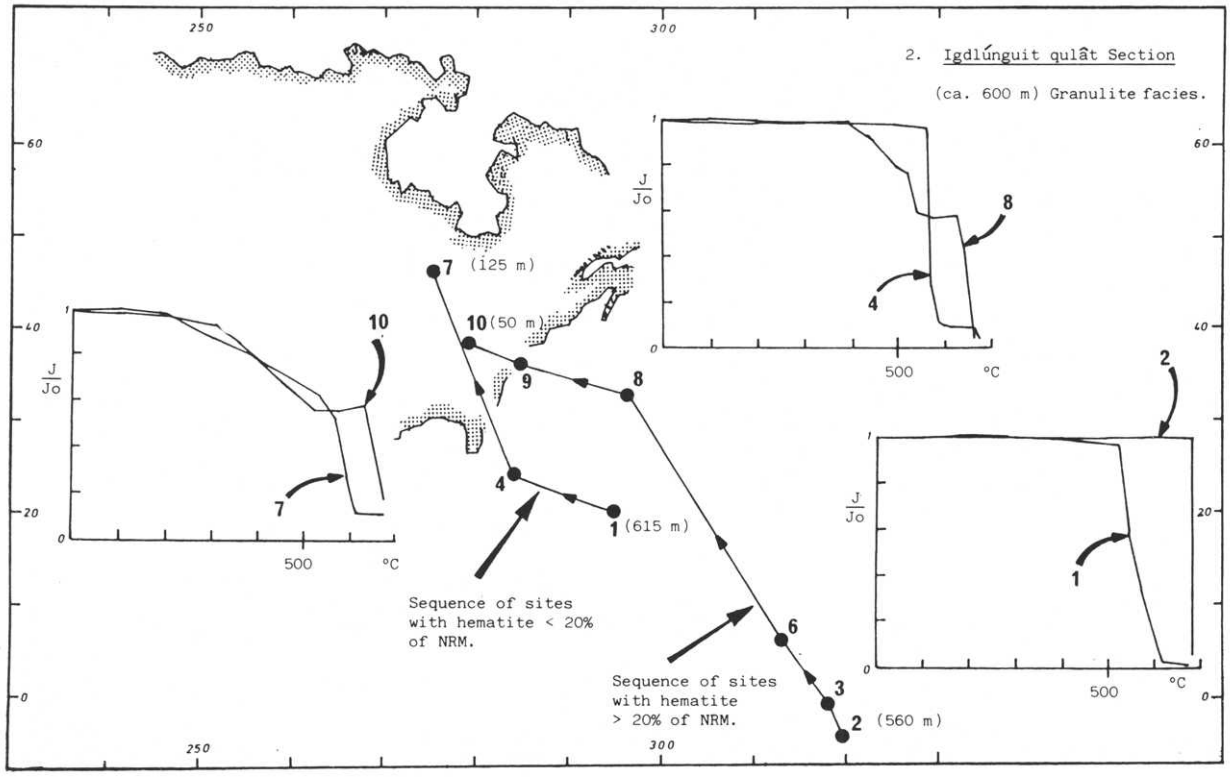


Fig. 7. The palaeopoles from section 2 renumbered as in Table 1 and plotted as a function of altitude. On the basis of blocking temperature characteristics they are plotted into two groups: those (2, 3, 6, 8, 9 and 10) with their remanence unblocked close to the Curie point of hematite and those (1, 4 and 7) with their remanence unblocked close to the Curie Point of magnetite. Some representative thermal demagnetisation curves are also illustrated.

tude sequences conform with the trend identified in the first section (Fig. 5). The systematic record strongly suggests that this area recorded separate a.p.w. motions during regional uplift and cooling depending on whether the predominant carrier was a higher (hemo-ilmenite) or lower (magnetite) blocking temperature phase.

3.4. Praestefjeldet section

This section utilises late Nag 2 diorite intrusions and all of these rock prove to have very high blocking temperature spectra (Fig. 8). All samples have more than 50% of their remanence still blocked at the Curie point of magnetite and some have 100% remaining at this stage. They also have directions of magnetisation contrasting with the other sections although there is a migration towards the directions of sections 1 and 2 in the site means going down this section (Fig. 8). Samples of opposing polarity are present at the lowest two sites and define this as a dipolar axis. There is no significant difference in the blocking temperature spectra between the samples falling in the groups of opposing polarities; this is unlike the clear difference present between sites 2, 3, 6, 8–10 and 1, 4 and 7 in section 2 and implies that this palaeofield reversal was a much shorter event than the a.p.w. motion identified in section 2. However, there is a small difference in the a.f. demagnetisation characteristics with all samples in the NE negative group exhibiting a small increase in remanence at about 15 mT; the SW positive group all have somewhat lower coercivities and exhibit a systematic fall in moment (Fig. 6). Similar behaviour to the former group is identified in the 'A' component of the Haliburton intrusions [19] of the Grenville Province where it is also attributed to a

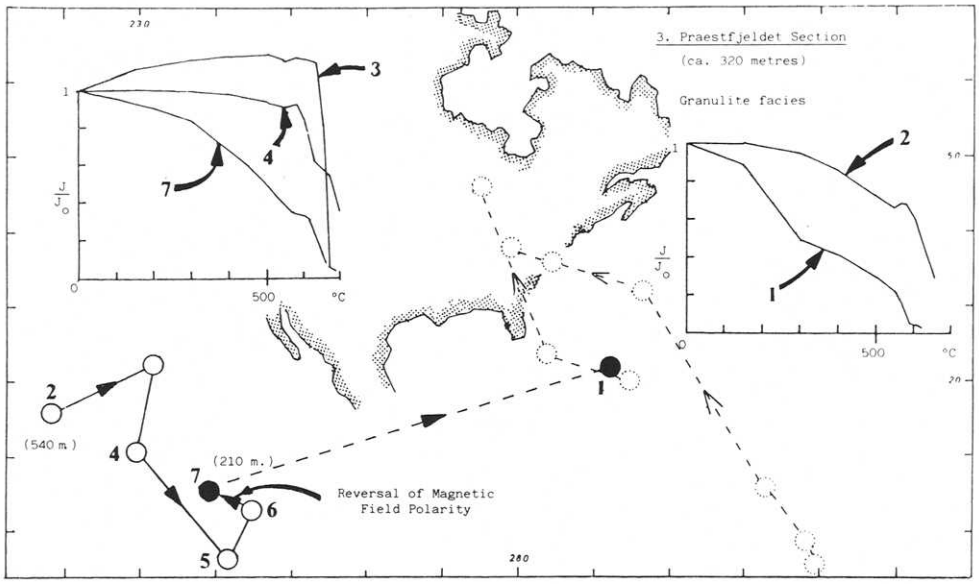


Fig. 8. The palaeopoles from section 3 renumbered as in Table 1 and plotted as a function of altitude together with some representative thermal demagnetisation curves. Site 1 is from country rock gneiss at the summit of this section; other sites are in diorites.

small antiparallel component. In this instance it suggests that the SW + magnetisation postdates the NW − direction. A site (39) in gneissose country rock at the top of section 3 has a lower blocking temperature spectrum and a Curie point attributed to magnetite; it yields a contrasting direction to the remaining sites with a palaeopole falling in the same field as that defined by sections 1 and 2.

On the south side of this fjord at Holsteinsborg Beckmann et al. [5] sampled five sites up the Mountain Kaellinghaetten to 620 m. All of these sites were in gneissose country rock; they have directions of magnetisation (corrected for anisotropy) which are close to site 1 of this section and have a group mean $D = 188°$, $I = 66°$. The demagnetisation spectra illustrated by these authors are all lower than the late Nag 2 diorites with 90% or more of the initial remanence unblocked close to the Curie point of magnetite.

3.5. Age of the magnetisation sequences

Only that part of the blocking temperature spectra above 300°C could have remained blocked since the time of cooling of these rocks [18] and so the maximum possible range for the acquisition of their stable remanence is about 300–680°C. The thermal demagnetisation spectra are generally taken to indicate a discrete or distributed range of blocking temperatures below the Curie point but cannot indicate the actual blocking temperatures because these will be lowered under conditions of slow uplift: the actual blocking temperatures will be those points at which the relaxation time (τ) becomes so long compared with the cooling time that the magnetisation can no longer follow the ambient field changes, and it is thus dependent on the rate of cooling. The relaxation time τ_1 at temperature T_1 can be related to the relaxation time τ_2 at temperature T_2 by the formula:

$$\frac{T_1 \ln (C\tau_1)}{J_{s_1} H_{c_1}} = \frac{T_2 \ln (C\tau_2)}{J_{s_2} H_{c_2}}$$

The activation energy, $\frac{1}{2} vJ_s H_c$, decreases to zero at the Curie point where all grains are unblocked whatever the time scale, and when the change of activation energy is not considered the change in blocking temperature with time is overestimated [20]. When changes in the activation energy are

taken into account the laboratory blocking temperature of magnetite is only slightly reduced from about 570 to 560°C [21]. The assessments [1,21] incorporating the variations of J_s and H_c with temperature indicate that the blocking temperature range is broadened, but the blocking temperature close to the Curie point is only slightly reduced, under conditions of slow cooling.

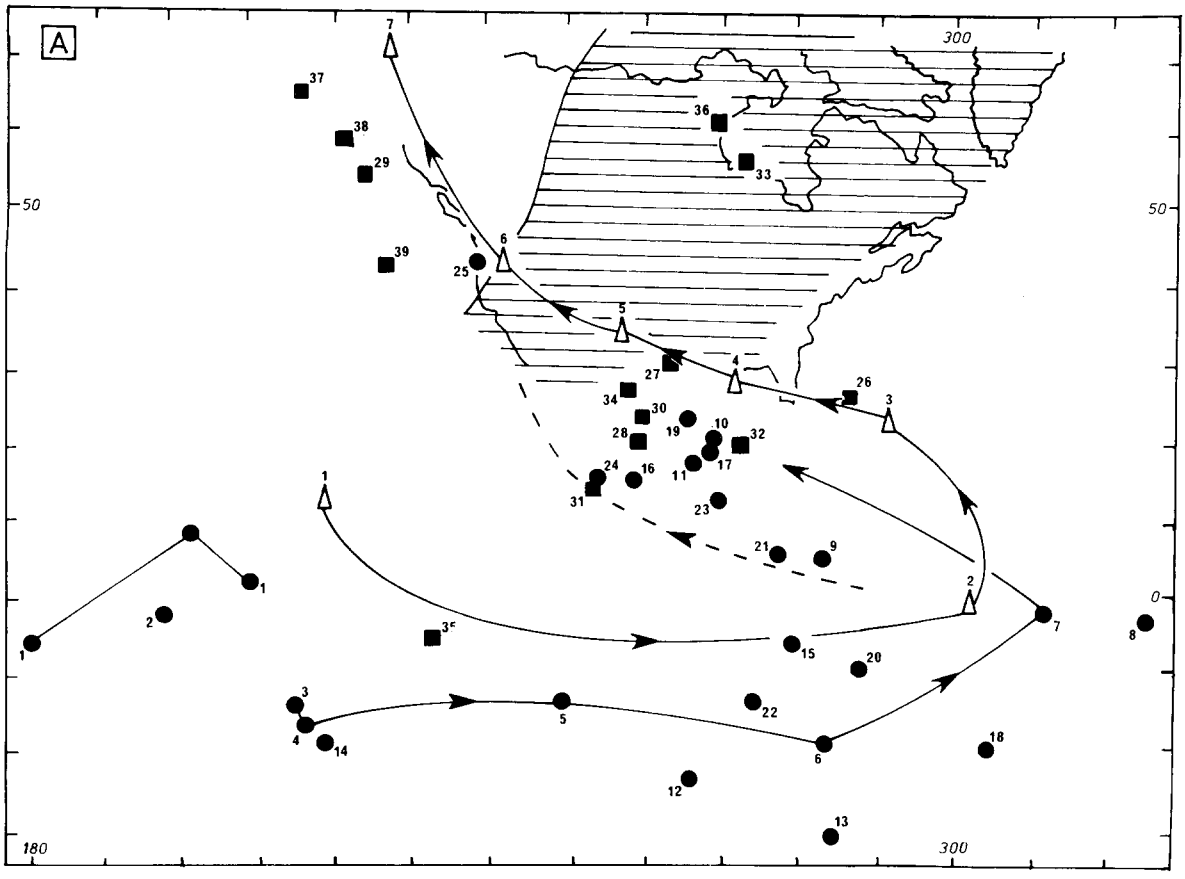
In the Nagssugtoqidian belt the uplift history is not yet documented in detail but pressures of 9.5 kbar were achieved during the peak of the Nagssugtoqidian metamorphism in the Holsteinsborg area [22]. In the analogous terrain of northwestern Scotland present erosion levels appear to have been at depths in excess of 10 km during the ca. 1800 Ma Laxfordian episode [23]. A later magnetisation at temperatures of 150–300°C during subsequent uplift would imply that the magnetisation was acquired at a depth of 8–15 km for a geothermal gradient of 20°C/km. Although these considerations give only a rough idea of the actual temperatures at which the magnetisation was acquired during slow cooling in any specific area, they provide generally grounds for examining these magnetisations in the context of K-Ar and Rb-Sr mineral ages, with the reservation that the temperatures which these minerals cease to act as open systems may also be dependent on the cooling rate [24].

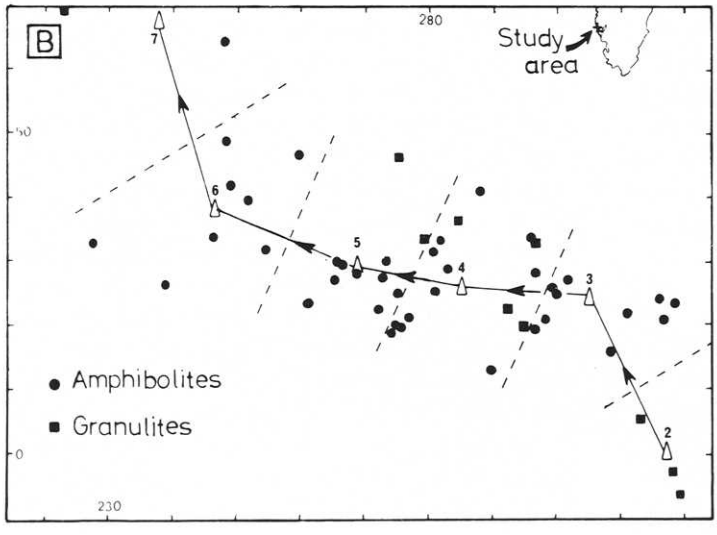
K-Ar biotite ages from the Nagssugtoqidian belt fall in the range 1790–1650 Ma [25] and are interpreted to represent the time that biotite closed to argon, i.e. ca. 200–300°C [26] and temperatures comparable to, or lower than, those at which the magnetisations were acquired. More definitive data relating to the area of the palaeomagnetic study are given in the form of K-Ar isochrons by Beckmann and Mitchell [27]; these authors derived isochron ages of 1700 ± 40 Ma and 1830 ± 20 Ma for gneiss samples in the vicinity of section 1 and they determined dolerite dyke whole-rock ages from the area ranging from 1720 ± 20 to 1890 ± 20 Ma. These determinations are from the granulite facies terrains and suggest that the magnetisations were acquired only shortly after the Nag 2 tectonic episode at ca. 1700–1670 Ma [14]; indeed, the late Nag 2 diorites of sections 2 and 3 may have a remanence dating from, or only shortly after, their time of intrusion. The age determinations may not, however, be entirely representative of the amphibolite facies terrain. Beckmann and Mitchell [27] derive a 1620 ± 50 Ma isochron from amphibolite facies gneisses at Sagdlerssuaq, 30 km NW of section 1 where the blocking temperature spectra of the characteristic magnetisations are also lower than in the granulite facies terrain ([27] and data of author).

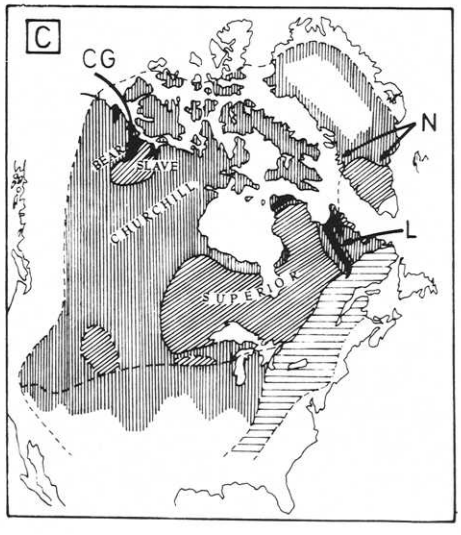
4. Correlations with the Laurentian Shield

In palaeomagnetic studies it is customary to compute mean palaeopoles representative of the collection for regional and intercontinental comparisons, and for inclusion in collated listing such as those of the Geophysical Journal. However, metamorphic terrains seldom record a single direction of magnetisation but instead (as in this study) indicate a migration of the palaeofield direction over a time interval. Representative mean poles have accordingly been computed for successive groups of data in Fig. 9 (inset) and Table 2 from the high blocking temperature group with E-W shallow magnetisations (section 3) and for poles falling within 20° subdividions of the SE → NW a.p.w. path (Figs. 5, 7 and 8). The collective poles define an a.p.w. swathe which is well-defined over approximately 100° of arc with the direction defined by the altitude sequence and the demagnetisation studies and which the collective age data suggest is about 1700 Ma in age. Only comparison with the range of dated poles from other Laurentian Shield rocks can improve the estimate of the time period represented by this path.

The relevant data are summarised in Fig. 9 and come from basement terrains mobilised during the ca. 1800 Ma Hudsonian episode and from sedimentary sequences in the upper part of the Coronation geosyncline which was infilled at ca. 2000–1700 Ma along the northwest margin of the Laurentian Shield [28]. These latter poles in stratigraphic sequence are 3–7 [29]. A secondary magnetic component [30] is very widespread in the Coronation and Labrador geosynclines (poles 8, 9, 10, 11 and 16) and it coincides very closely with the Greenland uplift magnetisations (poles 26–35). A Hudsonian overprint [31] is also present in the


A
300
50
0
180
300


B
280
230
Study area
Amphibolites
Granulites


C
CG
N
L
SLAVE
CHURCHILL
SUPERIOR

(> 2600 Ma) Superior craton (pole 1) where it was evidently acquired earlier in the mobile episode, and coincides with poles 3 and 4 from the Coronation geosyncline and the high blocking temperature component (pole 1) from the diorites of section 3 in West Greenland. This grouping of poles from different parts of the Laurentian Shield strongly suggests that poles 3 and 4 from the central part of the Coronation succession have not been subject to local rotation [29,30].

The a.p.w. trend (poles 2–7) from West Greenland substantiates the sequence of poles implied by the uppermost primary magnetisations and the secondary overprints in the Coronation ($6 + 7 \rightarrow 9 + 10 + 11$) and Labrador ($12 \rightarrow 15 + 16$) geosynclines. Morgan [2,4] also successfully defined a part of this trend (approximately between mean poles 2 and 5 of this study) by sampling at sea level across 20 km of the Nagssugtoqidian Front at the head of Itivdleq Fjord and 30 km to the west of section 1 (Fig. 1). It is noteworthy, however, that Morgan's trend (dashed line, Fig. 9) is displaced as the declinations become more westerly, to the SW of the trend defined by this study (dominated (Fig. 9 inset) by the data of section 1) and to the other side of the bulk of the North American and Greenlandic poles falling in this area. It is possible that this difference has a local tectonic significance and reflects a warping of the basement after the regional cooling and magnetisation.

The comparison with the North American data does not permit much refinement of the age estimate of the SE-NW a.p.w. trend in Fig. 9 because these poles have assigned ages ranging from 1800 to 1620 Ma; the age data relating to the Nag 2 episode imply that the former estimate may be up to 100 Ma too high. The youngest part of the swathe lies close to the Lewisian A2 magnetisations from northwest Scotland linked to K-Ar hornblende ages in the range 1650–1625 Ma, also to pole 37 from Ketilidian metavolcanics a metamorphic magnetisation assigned to uplift ages in the range 1700–1600 Ma, and finally to pole 38 from amphibolite at Sarfaquaqland near the centre of the Nagssugtoqidian belt with comparable magnetic properties to the Sagdlerssuaq rocks (pole 33) dated 1620 ± 50 Ma. New data [32,32] also define an extension of this a.p.w. path into the central Pacific via poles from northwest Scotland and the North West Territories of Canada which appear to be at least 1600 Ma in age but younger than poles illustrated in Fig. 9. Hence the path between poles 2 and 7 may represent as much as 100 Ma of a.p.w. but no more; this yields a minimum a.p.w. rate of 1°/Ma with a value of 2°/Ma being a more realistic estimate. In section 2 sites 2 and 3 fall on the same trend as sites 9 and 10 and all

Fig. 9. A. Palaeomagnetic poles from the Laurentian Shield relating to the interval ca. 1750–1650 Ma. North American poles (●) with assigned ages in brackets are (*=magnetisations specifically linked to the Hudsonian mobile episode): *1*=Nipissing diabase, "E" magnetisations*; *2*=Murdoch Thompson Lake and Menihek rocks (1855 ± 74); *3*=Tochatwi Formation (1873); *4*=Stark Formation (4); *5*=Takiyuak Formation (<4); *6*=Pearson Formation, "A" (<5); *7*=Et-Then Group primary; *8*=Pearson Formation, "B"; *9*=Pearson Formation, "C"; *10*=Coronation Group overprint*; *11*=Kahocella secondary*; *12*=Seward Formation, Labrador, "A" (1800); *13*=Retty peridotite, Labrador, "A"; *14*=Menihek Formation, Labrador, "A"; *15*=Menihek Formation, Labrador, "B"; *16*=Sokoman Iron Formation, Labrador, secondary*; *17*=metamorphosed Kaminak dykes (1892–1615); *18*=Flin-Flon, "A" (1800); *19*=Flin-Flon, "B" (1700–1600); *20*=Martin Formation (1635 ± 180); *21*=Dubawnt Group (1825 ± 4); *22*=Nonacho Group (2500–1700); *23*=Daly Bay metamorphics, "A" (1622 ± 30); *24*=Cape Smith metavolcanics (1650–1450); *25*=Wind River dykes (1880–1680). Greenlandic poles (■) corrected for Phanerozoic drift are: *26*=Kaellinghaetten Gneiss, Holsteinsborg; *27*=Nordre Strømfjord Gneiss; *28*=Kangamiut dykes south of Nagssugtoqidian Front; *29*=Kangamiut dykes north of Nagssugtoqidian Front; *30*=Godhåb rocks plus Amitsôq gneisses; *31*=Itivdleq dykes [2]; *32*=Itivdleq dykes [27]; *33*=Sagdlerssuaq dykes (1620); *34*=Angmagssalik gneisses (1800–1600); *35*=Nordfjord gneisses; *36*=Ketilidian metavolcanics (i) (1700–1600); *37*=Ketilidian metavolcanics (ii) (1700–1600); *38*=Sarfanguaqland amphibolites; *39*=Lewisian A2 magnetisations (1650–1625) form northwest Scotland and also corrected for Phanerozoic drift. The sequence of magnetisations derived from this study is indicated by the open triangles (*1–7*). The palaeopoles grouped together to derive these representative mean poles are indicated in the inset diagram (B). C. Outline of the Proterozoic geology of the Laurentian Shield showing the locations of the Coronation (*GC*) and Labrador geosynclines (*L*) and the Nagssugtoqidian mobile belt (*N*). Zones of inclined shading are terrains stabilised prior to 2300 Ma, zones of vertical shading are terrains mobilised during the ca. 1800 Ma Hudsonian events and the zone of horizontal shading is the ca. 1100 Ma Grenville mobile belt.

TABLE 2

Group mean directions of magnetisation and paleomagnetic poles defining the ca. 1700 Ma a.p.w. swathe (Fig. 9)

Pole reference No.	Site included in calculation	*N*	*D*	*I*	*R*	*k*	Palaeomagnetic pole				Pole position relative to North American corrected for Phanerozoic drift
							°N	°E	*dp*	*dm*	
1	34(i), 34(ii), 35, 36, 37, 38	6	75.6	−26.6	5.81	25.7	−7.4	55.6	7.9	14.6	39.0°E, 12.8°S
2	30(i), 32(i), 32(ii)	3	168.4	40.8	2.99	147.8	0.6	317.7	7.5	12.3	300.8°E, 0.4°S
3	1(i), 1(ii), 2(i), 2(ii), 3(ii), 11(i), 11(ii), 12(ii), 13(i), 13(ii), 18	11	182.1	65.1	10.96	276.1	24.0	305.5	3.6	4.5	291.1°E, 24.1°N
4	3(i), 4, 5(ii), 9(i), 9(ii), 10(i), 12(i), 15(ii), 19(i), 20, 21, 22, 27, 28, 31, 33(i), 39	17	209.5	64.8	16.86	115.7	25.8	285.0	4.3	5.4	270°E, 27.9°N
5	5(i), 6(i), 6(ii), 10(ii), 14(i), 14(ii), 15(i), 19(ii), 26, 29, 30	11	229.8	65.3	10.95	185.1	30.2	170.2	4.4	5.5	256.2°E, 33.6°N
6	24(i), 24(ii), 7(i), 7(ii), 8(i), 8(ii), 17(i)	8	260.5	64.0	7.92	83.0	37.8	246.5	7.8	9.7	232.0°E, 42.8°N
7	17(ii), 30(ii)	2	304.2	76.7	2.86	400.1	67.7	237.9	21.6	23.3	227.5°E, 73.0°N

have more than 50% of their total remanence unblocked at the Curie point of magnetite; they are separated by about 500 m in altitude and represent ca. 25–50 Ma of a.p.w. movement. This implies an uplift rate for this basement terrain of 10–20 m/Ma and comparable estimates are derived from the data of Fig. 5. This should be treated as a minimum value because of the generous estimate of the time period involved but it is broadly consistent with the geological evidence from mid-Proterozoic terrains [3]; in some of these mobile terrains where ca. 1700 Ma Rapakivi-type intrusions merge upwards into sub-volcanic associations, an uplift rate of up to 50 m/Ma seems to have applied, but this is still less than 20% of the likely uplift rate in the Caledonian orogenic belt [34] and more than an order of magnitude lower than Alpine rates. Dodson and McClelland-Brown [21] utilise their theoretical calculations in conjunction with the data of Morgan from this area to quantify some aspects of the regional uplift and cooling. In the present study sample directions in section 1 moved between 2 and 32° during the interval 550–570°C, one of the last stages of thermal cleaning here, but movements averaged about 9°. Their estimate of a polar movement of 17° during a cooling interval of 38°C is therefore entirely valid for the present study; since the cooling interval involved was only of the order of 25 Ma however, the cooling rate becomes about 2.5°C/Ma. Their estimate for the regional tilt across the Nagssugtoqidian Front is unaltered. It should be remembered that the exposed crustal levels were still at depths of the order of 10 km when the magnetisations were acquired, but the predominance of a post-Hudsonian remanence assigned to the interval < 1750 Ma and > 1600 Ma in terrains as widespread as northwest Scotland, Greenland, Labrador and the North West Territories of Canada [30–33] implies a discrete episode of crustal uplift beneath all of these terrains following the ca. 1800 Ma mobile episodes, and identifies magnetisation following one of the major "accretion superevents" [35] of Precambrian times.

5. Conclusions

These studies up three mountain sections in Greenland have shown that slowly-cooled basement terrains can yield a record of the sequential acquisition of magnetisations provided that the blocking temperature spectra of the magnetisations are taken into account. Some basement terrains have also yielded a lateral change in characteristic magnetisation [4,33] but they require a consideration of the late metamorphic history and mineral age data to be converted into an actual direction of a.p.w. motion; vertical sections are ideal for these studies because there is no ambiguity about the relative age of cooling. Collectively the data from basement terrains are providing the first realistic indication of the scale and characteristics of Proterozoic a.p.w. motions, and it is already quite clear that they are fundamentally different from Phanerozoic a.p.w. motions. Greenland is ideal for these studies because of the large exposed thickness of dissected terrain there, but with a judicious selection of locations it should be possible to define or refine the a.p.w. movements during uplift and cooling following the major Proterozoic (ca. 2100, 1800 and 1100 Ma) mobile episodes.

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